School of Computer Science Engineering and Technology

Lab No. - 19

Course-B. Tech.	**Type-** Core
Course Code- CSET301	**Course Name-** Artificial Intelligence and Machine Learning
Year- 2025	**Semester-** Odd
Date- 30-10-2025	**Batch-** 2023-2027

CO-Mapping

	CO1	CO2	CO3	CO4	CO5	CO6
Q1						

Lab – Implementation of PCA **Total Marks: 1**

Objective:

To implement PCA on the MNIST dataset using Scikit-learn to reduce dimensionality, visualize data clusters, and evaluate information retention and reconstruction error.

Problem Statement

The objective of this lab is to implement Principal Component Analysis (PCA) on the MNIST handwritten digit dataset using Scikit-learn. The MNIST dataset consists of 70,000 grayscale images of handwritten digits (0–9), each represented as a 28×28 image (784 features). Due to the high dimensionality of this dataset, PCA is applied to reduce the number of features while retaining as much of the original variance (information) as possible.

In this task, you will:

- **Perform data preprocessing (standardization) on the MNIST dataset.**
- **Apply PCA to reduce the dimensionality of the dataset to 2 components for visualization of data clusters.**
- **Reconstruct the original images from the PCA-reduced representation and evaluate the reconstruction error using Mean Squared Error (MSE).**
- **Interpret the results to understand the trade-off between dimensionality reduction and information loss.**

The goal is to develop a strong understanding of how PCA can be used for data compression, visualization, and noise reduction, and to analyze its effect on real-world image data.

Download the dataset from:

https://www.kaggle.com/c/digit-recognizer/data
(*Alternative source:*) tensorflow.keras.datasets.mnist

About Dataset:

The MNIST dataset contains grayscale images of handwritten digits (0 to 9), where each image is 28x28 pixels, resulting in 784 features. It's a benchmark dataset widely used in machine learning and computer vision.

Attribute Information:

- **Each sample: 784-pixel values (28x28 image flattened)**
- **Target: Integer digit label (0 to 9)**

Questions:

1. **Data Pre-processing:**
   - **Load the MNIST dataset using tensorflow.keras.datasets.**
   - **Flatten the 28x28 images to 784-length vectors.**
   - **Normalize pixel values between 0 and 1 using StandardScaler.**
2. **Apply PCA:**
   - **Use PCA to reduce dimensions from 784 to 2.**
   - **Display explained variance ratio for the top 2 components.**
   - **Comment on the total variance retained.**
3. **Visualization:**
   - **Plot the 2D PCA-transformed data using a scatter plot.**
   - **Use color to distinguish between digit classes (0–9).**
4. **Reconstruction:**
   - **Reconstruct the original images from the 2D components.**
   - **Calculate Mean Squared Error (MSE) between the original and reconstructed images.**